**11** A die in the form of a dodecahedron has its faces numbered from 1 to 12. The die is biased so that the probability that a score of 12 is achieved is different from any other score. The probability distribution of $X$, the score on the die, is given in the table in terms of $p$ and $k$, where $0 < p < 1$ and $k$ is a positive integer.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $P(X = x)$ | $p$ | $p$ | $p$ | $p$ | $p$ | $p$ | $p$ | $p$ | $p$ | $p$ | $p$ | $kp$ |

Sam rolls the die 30 times, Leo rolls the die 60 times and Nina rolls the die 120 times. They each plot their scores on bar line graphs.

**(a)** Explain whose graph is most likely to give the best representation of the theoretical probability distribution for the score on the die. **[1]**

**(b)** Find $p$ in terms of $k$. **[2]**

**(c)** Determine, in terms of $k$, the expected number of times Nina rolls a 12. **[3]**

**(d)** Given that Nina rolls a 12 on 32 occasions, calculate an estimate of the value of $k$. **[2]**

Nina rolls the die a further 30 times.

**(e)** Use your answer to part **(d)** to calculate an estimate for the probability that she obtains a 12 exactly 8 times in these 30 rolls. **[2]**

| | |
|---|---|
| **11(a)** | |
| **11(b)** | |
| **11(c)** | |
| **11(d)** | |

| | |
|---|---|
| **11(e)** | |
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